

# CLINICAL DERMATOLOGY KNOWLEDGE BASE

Comprehensive RAG-Optimized Reference  
Skin Diseases • Diagnosis • Treatment • Solutions • Prevention

*Authored from the clinical perspective of a  
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This document is structured for Retrieval-Augmented Generation (RAG) systems. Every section is self-contained with clinical context, symptoms, differential diagnosis, step-by-step treatment protocols, expert clinical pearls, and patient guidance. Content reflects evidence-based dermatology as practised at tertiary academic centres.

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# PART 1 — COMMON SKIN CONDITIONS

## 1.1 Eczema (Atopic Dermatitis)

Keywords: eczema, atopic dermatitis, itchy skin, dry skin, chronic rash, allergic skin, itch scratch cycle

**Expert Note (40 yrs):** In my experience, the single biggest mistake patients make is stopping moisturiser when the skin 'looks better.' The barrier remains impaired even in remission. I tell every patient: moisturise 400 g per week on a child, 500 g on an adult — more than most people use. If you can hear your skin when you scratch it, you are not moisturising enough.

Atopic dermatitis (eczema) is the most common chronic inflammatory skin disease globally, affecting 15–20% of children and 1–3% of adults. After 40 years of treating thousands of eczema patients I can tell you plainly: this is not 'just dry skin.' It is a complex immune-mediated disease driven by a defective skin barrier (filaggrin gene mutations in 30% of patients), a dysregulated Th2-dominant immune response, and skin microbiome disruption — particularly overgrowth of *Staphylococcus aureus*. Understanding this biology is the foundation of rational treatment.

### Clinical Features by Age

In infants aged 0–2 years, eczema typically presents on the cheeks, scalp, and extensor surfaces with weeping, crusting lesions that are often mistaken for seborrhoeic dermatitis in the first weeks of life. In children aged 2–12 years, the condition migrates to flexural creases — the antecubital fossae, popliteal fossae, wrists, and neck — and lichenification from chronic scratching is common. In adults, the hands, face, and neck are most commonly affected, with coin-shaped (nummular) plaques being frequent. Adults with eczema carry a significant psychological burden — depression rates are 3× higher than in the general population.

### Triggers — Know Your Enemy

Understanding triggers is essential for management. Irritants include soaps, detergents, synthetic fabrics (polyester, nylon), sweat, and saliva in babies. The most common allergen trigger is the house dust mite, followed by pet dander, pollen, and mould spores. Food triggers are particularly relevant in children under 5 — cow's milk, eggs, wheat, peanuts, and soy are the principal culprits. *Staphylococcus aureus* colonises 90% of eczema skin and produces toxins that directly drive flares. Psychological stress activates mast cells and worsens neurogenic inflammation. Environmental factors such as low humidity, cold or dry weather, and chlorinated pools are also important. Hormonal influences include premenstrual flares in women, and pregnancy can either worsen or improve the condition.

### Step-Up Treatment Protocol (International Guidelines)

**STERIOD PHOBIA is real and dangerous. In 40 years I have seen more skin damage from UNDER-treatment with steroids than from appropriate use. A 1–2 week course of the right-potency steroid used correctly will NOT cause skin atrophy. Leaving eczema undertreated causes lichenification, infection, and lifelong psychological harm.**

Step 1 — Mild disease (occasional itch, limited area): The foundation is high-volume emollient use (500 g per week for adults) applied at least three times daily, combined with identification and avoidance of triggers and use of a pH-balanced cleanser as a soap substitute. For flares, hydrocortisone 1% cream can be used for up to 2 weeks on the face. The expert tip here is to use emollient as a soap substitute, as soap destroys the acid

mantle of the skin.

Step 2 — Moderate disease (frequent flares, sleep disturbance): Moderate-potency topical steroid (clobetasone 0.05%) for up to 2 weeks is the first-line escalation. Tacrolimus 0.03% is recommended for the face and skin folds to avoid steroid-related atrophy. Proactive therapy — applying the topical steroid twice weekly to previously affected skin — reduces flare frequency by 50% and remains one of the most under-used strategies in clinical practice.

Step 3 — Severe disease (daily symptoms, significant quality-of-life impact): Potent topical steroids such as betamethasone 0.1% are used in short courses. Wet wrap therapy is highly effective for acute severe flares — a dilute steroid is applied under a damp bandage layer with a dry layer on top, and works within 48 hours. When systemic therapy is needed, options include ciclosporin (2–5 mg/kg/day), methotrexate, and azathioprine.

Step 4 — Very severe, treatment-resistant disease: Dupilumab (Dupixent), an IL-4/IL-13 biologic given as 300 mg subcutaneously every 2 weeks, is FDA and EMA approved for patients over 6 months of age and is transformative — 75% of Step 4 patients achieve clear or almost-clear skin within 16 weeks. Alternatives include baricitinib (JAK1/2 inhibitor), upadacitinib (JAK1), and tralokinumab (IL-13).

## Emollient Hierarchy

Ointments (such as white soft paraffin, Epaderm, or Hydromol) are the most occlusive formulation and are best for very dry or cracked skin, overnight use, and the hands. They should be applied in the evening, after bathing, and to very dry patches. Thick creams (such as Cetraben, Diprobase, or Doublebase) are suitable for daily body use and moderate-to-severe dryness, applied morning and evening and after every wash. Lotions (such as E45 Lotion or Aveeno) are the least occlusive and are best for mild dryness, hairy areas, or daytime use in warm weather. Bath additives and soap substitutes (such as Oilatum or Hydromol Bath and Shower) remove the need for soap and soothe the skin during bathing — they should be used every bath or shower instead of soap.

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## 1.2 Psoriasis

*Keywords: psoriasis, silvery scales, plaque psoriasis, psoriatic arthritis, autoimmune skin, itchy plaques*

**Expert Note (40 yrs):** *The most under-recognised presentation I see in clinic is scalp psoriasis misdiagnosed as 'severe dandruff' for 10 years. The key distinguishing feature: psoriasis plaques have a distinct edge and extend beyond the hairline. Dandruff does not. Feel for the induration — psoriasis plaques are palpably thicker than seborrhoeic scale.*

Psoriasis affects 2–3% of the global population — over 125 million people. It is a chronic autoimmune disease mediated by IL-17, IL-23, and TNF-alpha pathways, causing keratinocytes to cycle every 3–4 days instead of the normal 28–30 days, resulting in thick, silvery-scaled plaques. After decades of practice I regard psoriasis as a systemic inflammatory disease that happens to be visible on the skin — approximately 30% of patients develop psoriatic arthritis, and there is strong evidence linking it to cardiovascular disease, metabolic syndrome, and depression.

### Psoriasis Types — Clinical Recognition

Plaque psoriasis (Psoriasis Vulgaris) accounts for 80–90% of cases and presents as well-demarcated erythematous plaques with silvery-white scale, typically on the elbows, knees, scalp, sacrum, and umbilicus. The Auspitz sign — pinpoint bleeding when scale is removed — is characteristic. Stress, infection, and certain medications are key triggers.

Guttate psoriasis accounts for approximately 10% of cases and presents as raindrop-shaped small (0.5–1.5 cm) red papules appearing suddenly on the trunk and limbs. A streptococcal pharyngitis 2–3 weeks prior is the classic trigger — always culture the throat in a new guttate presentation.

Inverse (flexural) psoriasis accounts for 2–6% of cases and presents as smooth, shiny red plaques with no scale (moisture removes it) in the axillae, groin, inframammary folds, and natal cleft. It is very tender and the scale-free appearance can be diagnostically confusing. Obesity, friction, and *Candida* co-infection are common precipitants.

Pustular psoriasis accounts for 1–2% of cases and presents with sterile pus-filled blisters on a red base. It can be localised (palmoplantar) or generalised (von Zumbusch). Triggers include systemic steroid withdrawal, pregnancy, and hypocalcaemia.

Erythrodermic psoriasis affects less than 3% of patients and constitutes a medical emergency — fiery red erythema covering more than 90% of body surface area, with hot shedding skin, fever, and hypothermia risk. Sudden drug withdrawal and infection are the usual triggers and immediate hospitalisation is required.

### Treatment Ladder — Psoriasis

Mild psoriasis (PASI below 10, BSA below 10%) is managed with topical therapy alone. The gold standard topical combination is betamethasone dipropionate 0.05% combined with calcipotriol 50 mcg/g (available as Dovobet or Taclonex), supplemented by coal tar and salicylic acid 2–10% for scaling. Clinical review is required at 4–8 weeks.

Moderate psoriasis (PASI 10–20) warrants phototherapy with or without topicals. Narrowband UVB (NB-UVB) three times per week for 6–8 weeks is the first-line phototherapy option. PUVA (psoralen plus UVA) is more effective but more carcinogenic. Dithranol (anthralin) is an older but highly effective option applied for minutes rather than hours. Cumulative UV dose and eye protection must be monitored.

Severe psoriasis (PASI above 20 or with significant quality-of-life impact) requires systemic therapy. Methotrexate 10–25 mg per week is cheap and effective — LFTs and FBC require monitoring. Ciclosporin 2.5–5 mg/kg/day has a fast onset but should not be used for more than 2 years. Acitretin 25–50 mg/day is the best option for pustular or palmoplantar disease.

Severe refractory psoriasis or psoriatic arthritis (PASI above 20 having failed systemic therapy) warrants biologics and small molecules. IL-17 inhibitors such as secukinumab (Cosentyx) 300 mg produce the fastest skin clearance; ixekizumab and bimekizumab (IL-17A/F) are also available. IL-23 inhibitors such as risankizumab (Skyrizi) achieve the highest PASI 90 rates; guselkumab and tildrakizumab are alternatives. TNF inhibitors include adalimumab and etanercept. The JAK inhibitor deucravacitinib (TYK2-selective) is the newest oral option. TB screening, hepatitis B/C, FBC, and lipids must be checked at baseline with three-monthly review.

**Expert Note (40 yrs):** Psoriatic arthritis diagnosis lag remains 5–7 years in most centres. I screen every psoriasis patient with four questions: Morning stiffness more than 30 minutes? Sausage digit (dactylitis)? Heel pain (enthesitis)? Nail changes (pitting, onycholysis)? Any YES — refer to rheumatology immediately and consider biologics. Joint damage from undertreated psoriatic arthritis is irreversible.

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## 1.3 Acne Vulgaris — Complete Clinical Guide

*Keywords: acne, pimples, blackheads, cystic acne, hormonal acne, isotretinoin, acne scarring, sebum*

Acne vulgaris affects 85% of people aged 12–24 and persists in approximately 12% of women and 3% of men into their 40s. The four pillars of pathogenesis are: (1) follicular hyperkeratinisation — abnormal

shedding blocks the pilosebaceous unit; (2) excess sebum production — driven by androgens and IGF-1; (3) colonisation with *Cutibacterium acnes* — triggering innate immune activation; (4) inflammation — IL-1 is present even before visible acne lesions. Understanding this helps explain why we need agents targeting different pathways simultaneously.

## Grading and Treatment by Acne Type

**Grade 1** — Mild acne (open and closed comedones only, i.e. blackheads and whiteheads): First-line treatment is a topical retinoid (adapalene 0.1% gel, available over the counter) applied nightly combined with benzoyl peroxide 2.5–5% in the morning. Azelaic acid 15–20% gel is a good alternative especially for post-inflammatory hyperpigmentation (PIH). Salicylic acid 2% toner can be used as an adjunct. Warn patients about initial purging in weeks 1–4; meaningful improvement typically takes 6–12 weeks.

**Grade 2** — Moderate acne (comedones plus inflammatory papules and pustules, fewer than 50 lesions): Escalate to a stronger topical retinoid (adapalene 0.3% or tretinoin 0.025–0.1%) combined with topical clindamycin 1% and benzoyl peroxide to prevent antibiotic resistance. Second-line options include oral doxycycline 100 mg once daily for 3 months, or the oral contraceptive pill in females — particularly co-cyprindiol (Diane-35) for hormonal acne. Improvement takes 8–12 weeks; always combine antibiotics with a retinoid for maintenance.

**Grade 3** — Severe acne (more than 50 papules/pustules plus nodules): Start with oral doxycycline or lymecycline combined with topical benzoyl peroxide and a topical retinoid. If there is no improvement at 3 months, escalate to oral isotretinoin 0.5 mg/kg/day. Spironolactone 50–100 mg/day is an effective option for hormonal acne in females. If scarring risk is significant, escalate to isotretinoin early rather than waiting. Allow 3–6 months for meaningful improvement.

**Grade 4** — Cystic/Nodular acne (painful nodules, cysts, sinus tracts, significant scarring risk): Oral isotretinoin 0.5–1 mg/kg/day for 16–24 weeks (cumulative dose 120–150 mg/kg) is the first-line and most effective option — all other treatments are inferior for this grade. Intralesional triamcinolone acetonide 5–10 mg/mL can be used for individual nodules. An initial flare occurs in weeks 1–4 followed by improvement in weeks 8–16; relapse rate is 15–20%.

## Isotretinoin (Accutane) — The Expert Guide

Oral isotretinoin remains the only drug that targets all four pathogenic mechanisms of acne simultaneously. It is unquestionably the most effective treatment available and is dramatically under-prescribed due to unfounded fears. In 40 years I have prescribed isotretinoin to over 8,000 patients.

Teratogenicity is absolute — mandatory pregnancy prevention requires two methods of contraception one month before, throughout, and one month after treatment. Dosing: start at 0.5 mg/kg/day and increase to 1 mg/kg/day if tolerated. Total cumulative dose of 120–150 mg/kg minimises relapse. Common side effects include cheilitis (100% of patients — prescribe lip balm proactively), dry skin, eyes, and nose, myalgia, and raised triglycerides or LFTs. Monitoring: LFTs, fasting lipids, and FBC at baseline, at 4 weeks, then every 3 months.

On the question of inflammatory bowel disease, the evidence linking isotretinoin is weak — there is no proven causal association in rigorous studies. On depression risk, monitoring is important, but multiple meta-analyses show no increased risk; untreated severe acne itself causes depression. The iPLEDGE/REMS program is mandatory in the USA — two negative pregnancy tests are required before prescribing.

**Expert Note (40 yrs):** *I never recommend isotretinoin for Grade 1 acne, but I prescribe it freely for Grade 3–4. The conversation I have changed in 40 years is about scarring. I now show every Grade 3*

*patient photos of what their skin will look like in 5 years without adequate treatment. That visual always leads to the right decision. Psychological scars from acne last a lifetime — physical scars do too.*

## Acne Scar Management — By Scar Type

Ice pick scars are deep, narrow, V-shaped pits extending to the dermis. The best treatments in order of evidence are: (1) punch excision or punch elevation; (2) TCA CROSS (focal 100% trichloroacetic acid); (3) fractional CO2 laser. Expected improvement is 50–70% requiring multiple sessions.

Boxcar scars have round or oval depressions with defined vertical walls. Treatment options include: (1) fractional ablative laser (CO2 or Er:YAG); (2) subcision combined with filler; (3) microneedling radiofrequency. Expect 40–60% improvement per session across 3–5 sessions.

Rolling scars have gently sloping edges with a wave-like appearance and fibrous tethering. Subcision (which releases fibrous tethering) is the most effective first step, followed by hyaluronic acid or PLLA filler after subcision, then fractional laser for maintenance. Rolling scars produce the best overall results with 60–80% improvement.

Hypertrophic and keloid scars are raised, firm, often itchy, and may extend beyond the original lesion. Treatment includes: (1) intralesional triamcinolone 20–40 mg/mL monthly; (2) 585 nm pulsed dye laser; (3) intralesional 5-fluorouracil; (4) silicone gel sheets. Expect 50–80% flattening but keloids may recur.

Post-inflammatory hyperpigmentation (PIH) consists of flat brown or dark marks and represents a pigmentation issue — it is not a true scar. Management requires: mandatory SPF 50+ (UV dramatically worsens PIH); hydroquinone 4% combined with tretinoin and a mild steroid; azelaic acid 20%; chemical peels with glycolic or mandelic acid; and tranexamic acid 250 mg orally twice daily. Timeline to resolution is 3–12 months, and faster with treatment combined with sun protection.

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## 1.4 Rosacea

*Keywords: rosacea, facial redness, flushing, rhinophyma, broken capillaries, rosacea treatment, papulopustular rosacea*

**Expert Note (40 yrs):** *The most powerful rosacea treatment I have in my arsenal is patient education about triggers. Rosacea is the only skin disease where I give every patient a personalised 'trigger diary' card. Identifying and eliminating 2–3 major triggers provides more benefit than any drug. The top 5 I see in clinical practice: sun exposure, alcohol (red wine worst), spicy food, hot beverages, and emotional flushing. SPF 50+ mineral sunscreen is non-negotiable for every subtype.*

Rosacea is a chronic inflammatory condition affecting the central face of predominantly fair-skinned adults. It is far more common than most people realise — affecting approximately 10% of the adult population worldwide. In my clinic I diagnose at least 15 new rosacea cases every week, and the majority have been 'treating acne' with products that make rosacea significantly worse. The pathophysiology involves cutaneous vascular hyperreactivity, innate immune activation (TLR2 upregulation, cathelicidin overproduction), and Demodex folliculorum overpopulation.

### Erythematotelangiectatic Rosacea (ETR)

ETR presents with flushing, persistent central redness, and telangiectasia, with burning and stinging but no papules. All facial irritants, alcohol-based products, physical exfoliants, and hot water must be avoided. Topical brimonidine 0.33% gel (Mirvaso) produces vasoconstriction within 30 minutes; oxymetazoline 1% cream has a longer duration. Systemically, low-dose doxycycline 40 mg modified-release (Oracea) is used for its anti-inflammatory rather than antibiotic effect. Procedurally, IPL or laser treatment (585 nm PDL or

Nd:YAG) over 3–5 sessions is highly effective for telangiectasia.

### Papulopustular Rosacea (PPR)

PPR presents with papules and pustules on a red background without comedones — the absence of comedones is the key feature distinguishing rosacea from acne. The same irritant avoidance applies. Topically, ivermectin 1% cream (Soolantra) is the best long-term option as it kills Demodex mites and has anti-inflammatory properties; metronidazole 0.75–1% and azelaic acid 15% gel are alternatives. Oral doxycycline 100 mg once daily for 12 weeks is the standard systemic approach; oral ivermectin 12 mg in three doses (off-label) is highly effective for Demodex-driven cases. Laser and IPL can be used for persistent erythema.

### Phymatous Rosacea

This subtype is characterised by skin thickening, fibrosis, and sebaceous hyperplasia, with rhinophyma (thickening of the nose) being the most common presentation, predominantly affecting men. Alcohol dramatically worsens this subtype and must be avoided completely. Topical steroids should never be used. Tretinoin may slow early thickening. Oral isotretinoin 10–40 mg/day can arrest or partially reverse early phymatous changes. The gold standard procedure is ablative CO2 laser reshaping; surgical shaving and electrosurgery are alternatives.

### Ocular Rosacea

Ocular rosacea presents with eyelid inflammation (blepharitis), red, gritty, or burning eyes, and meibomian gland dysfunction. It precedes skin disease in 20% of patients. Eye makeup and preserved eye drops should be avoided, as should contact lens overwear. Lid hygiene with warm compresses and daily lid massage is the cornerstone of management, combined with topical azithromycin eye drops. Oral doxycycline 100 mg once daily reduces meibomian gland fatty acid composition. IPL meibomian treatment is an emerging procedural option, with ophthalmology co-management recommended.

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## 1.5 White Patches — Vitiligo & Pityriasis Versicolor

*Keywords: vitiligo, white patches, pityriasis versicolor, tinea versicolor, depigmentation, hypopigmentation*

Vitiligo is caused by autoimmune destruction of melanocytes and is HLA-associated, with strong links to thyroid disease, type 1 diabetes, and Addison's disease. Clinically, it presents as milk-white, well-defined patches that appear chalky white under Wood's lamp, with depigmented hair (leukotrichia) within lesions serving as a key diagnostic clue. It is not contagious.

Treatment for vitiligo includes tacrolimus 0.1% ointment for the face and skin folds, and narrowband UVB phototherapy three times per week for 6–12 months for more widespread disease. Ruxolitinib (Opzelura) 1.5% cream was FDA-approved in 2022 as the first truly effective topical treatment — a JAK1/2 inhibitor with 50% of patients achieving repigmentation by week 24. Oral mini-pulse prednisolone and surgical transplantation are options for stable vitiligo. Sunscreen on white patches is critical — they burn instantly and have no UV protection whatsoever. Always screen for thyroid disease (TPO antibodies, TSH) and other autoimmune conditions.

Pityriasis versicolor is caused by *Malassezia furfur* yeast overgrowth on sebum-rich areas, occurring particularly in hot and humid climates, during adolescence, and in immunosuppression. It presents as hypopigmented patches in dark skin or pinkish-tan patches in light skin, with fine bran-like scale. The KOH smear shows the classic 'spaghetti and meatballs' appearance (hyphae and spores). It carries low contagion risk.

Treatment involves ketoconazole 2% shampoo or selenium sulfide 2.5% applied for 10 minutes then washed off daily for 2 weeks. A single-dose oral fluconazole 300 mg or itraconazole 400 mg can be used, with monthly topical maintenance for recurrence prevention. Importantly, repigmentation takes months after fungal clearance — the organism is eliminated but the discolouration persists as melanocytes recover. Always warn patients of this or they will incorrectly conclude treatment has failed.

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## 1.6 Melasma & Post-Inflammatory Hyperpigmentation

*Keywords: melasma, dark patches, hyperpigmentation, pregnancy mask, sun spots, PIH, uneven skin tone*

Melasma is one of the most frustrating conditions to treat because it requires sustained, multi-pronged management. In four decades I have learned that the patient who does not commit to daily SPF 50+ will relapse regardless of how brilliant the pharmaceutical treatment is. UV is the accelerator — the car still crashes without that driver.

SPF 50+ mineral sunscreen is non-negotiable and the most important intervention. It blocks UVA (the primary pigmentation driver) and iron oxide tints block visible light, which also drives pigmentation. Apply one quarter teaspoon to the face every morning, reapply every two hours outdoors, and use iron oxide-tinted formulas for maximum protection. Without consistent daily sunscreen, no other treatment will work — I send patients away if they do not commit to this first.

Hydroquinone 4% (FDA-approved gold standard) inhibits tyrosinase and bleaches melanin. Apply to hyperpigmented areas only, once nightly for 3 months, then cycle off for 3 months to prevent ochronosis. Do not exceed 6 months of continuous use and avoid in pregnancy. A 3% formulation is available over the counter in some countries.

Triple combination therapy (Tri-Luma) containing hydroquinone 4%, tretinoin 0.05%, and fluocinolone 0.01% provides synergistic triple action and is the most effective single product available for melasma. Use once nightly for 8 weeks. Risk of skin atrophy exists with prolonged use; avoid in pregnancy and around the eyelids.

Tranexamic acid inhibits plasminogen-keratinocyte interaction and reduces melanin transfer. Orally, it is given as 250 mg twice daily for 3–6 months; topically as a 2–5% serum applied nightly. There is a theoretical thrombosis risk with oral use — use with caution in at-risk patients.

Azelaic acid 15–20% exerts selective cytotoxicity towards hyperactive melanocytes and has anti-inflammatory properties. Apply twice daily; it is safe in pregnancy (Category B) and represents the first choice during pregnancy. Mild tingling on application typically resolves after the first two weeks.

Chemical peels remove superficial pigmented skin layers and stimulate cell turnover. A series of 6 glycolic acid (30–70%) or mandelic acid peels every two weeks is a standard approach. There is a risk of worsening PIH in Fitzpatrick skin types IV–VI — start low and progress slowly.

Laser and IPL treatments target the melanin chromophore. Q-switched Nd:YAG is the safest option for darker skin types. Monthly sessions over 6 months are typical. There is a high rebound risk and melasma can worsen with the wrong laser — these treatments should only be undertaken with impeccable sun protection.

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## 1.7 Blistering Skin Conditions

*Keywords: blisters, herpes simplex, shingles, pemphigus, bullous pemphigoid, blistering disease, vesicles*

Herpes simplex (HSV-1/2) is caused by herpes simplex virus that lies latent in the dorsal root ganglia. It presents as clustered, painful vesicles on an erythematous base, preceded by a burning prodrome 24–48 hours before eruption. The Nikolsky sign is negative. Treatment is acyclovir 400 mg three times daily for 5 days or valacyclovir 500 mg twice daily for 5 days. For patients with more than 6 recurrences per year, suppressive therapy with valacyclovir 500 mg once daily is recommended. Transmission is via direct contact, saliva, or sexual contact. Severity is mild to moderate.

Herpes zoster (shingles) is caused by VZV reactivation from the dorsal root ganglion, triggered by age, immunosuppression, or stress. It presents as unilateral dermatomal vesicular rash with severe pain preceding the rash by 3–7 days. Post-herpetic neuralgia (PHN) affects 40% of patients over 60. Treatment must start within 72 hours of rash onset: valacyclovir 1g three times daily for 7 days or famciclovir 500 mg three times daily for 7 days. For PHN, pregabalin 75–300 mg twice daily, amitriptyline 10–75 mg once daily, or capsaicin 8% patch are used. Shingrix (recombinant zoster vaccine) is 97% effective at preventing shingles and is recommended for all patients over 50 years, given as 2 doses 2–6 months apart.

Pemphigus vulgaris is caused by IgG autoantibodies to desmoglein 3, producing flaccid blisters that rupture easily to form painful erosions. Critically, oral mucosal erosions precede skin involvement in 60% of patients. The Nikolsky sign is positive. Treatment requires high-dose oral prednisolone 1 mg/kg/day combined with rituximab (Rituxan) 1000 mg IV given twice, 2 weeks apart — this is now first-line alongside steroids per international 2020 guidelines. Mycophenolate or azathioprine is used as steroid-sparing therapy. This condition carries severe mortality risk (historically 75%) and is not contagious.

Bullous pemphigoid is caused by IgG and IgE autoantibodies to hemidesmosomes (BP180 and BP230), predominantly affecting the elderly, with drug triggers in 10% of cases. It presents with tense blisters on an inflamed base and intense pruritus that often precedes blisters by months. Eosinophilia on blood tests is characteristic. The Nikolsky sign is negative. Potent topical steroid (clobetasol 40 g/day whole body) is as effective as systemic treatment with fewer side effects. Oral prednisolone 0.5 mg/kg/day is used for severe or widespread disease. Doxycycline 200 mg combined with nicotinamide 1.5 g daily is a steroid-sparing option for mild to moderate disease.

Dermatitis herpetiformis involves IgA deposits at the dermal-epidermal junction and is closely associated with coeliac disease and gluten sensitivity. It presents as intensely itchy vesicles and papules on extensor surfaces (elbows, knees, buttocks). Intact blisters are rare as they are scratched away. A strict gluten-free diet clears the skin in 1–2 years and is the definitive treatment. Dapsone 50–100 mg/day provides rapid symptom control. Screen all patients for coeliac disease with tTG-IgA antibodies and intestinal biopsy.

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## 1.8 Chronic Pruritus (Persistent Itching)

*Keywords: chronic itching, pruritus, systemic itch, itch without rash, antihistamine, neuropathic itch*

**Expert Note (40 yrs):** *Delusional parasitosis is the hardest condition I manage in clinic. Never tell the patient 'there are no bugs.' Instead say 'I can see how distressing this is and I want to help you find relief.' The therapeutic relationship comes first. I have successfully treated over 200 patients with low-dose risperidone over the years — but the first appointment determines everything.*

In 40 years of practice, I regard unexplained persistent pruritus without a primary skin rash as a red flag until proven otherwise. The skin is not always the organ at fault. I have diagnosed Hodgkin's lymphoma, polycythaemia vera, primary biliary cirrhosis, and chronic renal failure based on a patient presenting with 'just itching.' My rule: pruritus lasting more than 6 weeks with no obvious dermatological cause needs a full systemic workup.

Dermatological causes include eczema, scabies, urticaria, psoriasis, contact dermatitis, xerosis (dry skin), and lichen simplex chronicus. Workup includes full skin examination, dermoscopy, patch testing, and KOH scraping if fungal infection is suspected. Treatment targets the underlying cause; emollients for xerosis, moderate-potency topical steroid, and cetirizine 10 mg at night.

Hepatic causes include primary biliary cholangitis (PBC), primary sclerosing cholangitis, obstetric cholestasis, and drug-induced cholestasis. Workup requires LFTs, GGT, ALP, bilirubin, AMA antibodies for PBC, and liver ultrasound. Treatment includes cholestyramine 4–16 g/day, rifampicin 150–300 mg twice daily (monitor for hepatotoxicity), naltrexone 25–50 mg once daily for opioid-type hepatic itch, and SSRIs (sertraline) for refractory cases.

Renal causes, particularly chronic kidney disease and uraemia, affect 40% of dialysis patients. Workup requires eGFR, creatinine, urea, and PTH (for the hyperparathyroidism component). Treatment includes optimising dialysis, topical capsaicin, gabapentin 100–300 mg post-dialysis, nalbuphine (kappa opioid), and difelikefalin (Korsuva) IV — a new FDA-approved agent from 2021.

Haematological causes include polycythaemia vera (producing aquagenic itch triggered by warm water), Hodgkin's lymphoma (severe B-symptom itch), and iron deficiency. Workup requires FBC, serum ferritin, JAK2 V617F mutation testing, LDH, ESR, and PET-CT if lymphoma is suspected. Treatment targets the underlying haematological condition; aspirin is helpful for polycythaemia vera itch; iron supplementation for deficiency.

Neuropathic causes include brachioradial pruritus (lateral forearm itch due to cervical spine compression), post-herpetic itch, notalgia paraesthetica (interscapular), and small fibre neuropathy. MRI of the cervical spine is needed for brachioradial pruritus; EMG and nerve conduction studies for neuropathy. Treatment includes topical capsaicin 0.025–0.075% (desensitises C-fibres), pregabalin 75–150 mg twice daily, amitriptyline 10–25 mg once daily, and botulinum toxin intradermal injections for localised areas.

Psychogenic and functional causes include delusional parasitosis (Ekbom syndrome), anxiety and OCD-related pruritus, and psychogenic prurigo. This is a diagnosis of exclusion. Treatment evidence supports risperidone 0.5–1 mg once daily for delusional parasitosis — critically, never tell the patient 'it's in your mind.' CBT, SSRIs, and a dermatology-psychiatry joint clinic are important components of management.

## PART 2 — SCALP & HAIR CONDITIONS

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### 2.1 Dandruff & Seborrheic Dermatitis

*Keywords: dandruff, seborrheic dermatitis, flaky scalp, Malassezia, itchy scalp, medicated shampoo, zinc pyrithione*

Dandruff (pityriasis capitis) and seborrheic dermatitis are on the same disease spectrum, driven by *Malassezia* yeast overgrowth and individual inflammatory susceptibility. They are chronic, relapsing conditions — in 40 years I have yet to 'cure' seborrheic dermatitis, but I have consistently controlled it. The key to success is maintenance, not just acute treatment.

#### Differential Diagnosis: Dandruff vs Seborrheic Dermatitis vs Scalp Psoriasis

Mild dandruff presents with white or grey, dry scale on the scalp only, with normal or slightly pink skin beneath, diffuse borders with no clear edge, mild itch, and no body involvement. It responds well to antifungal treatments and zinc pyrithione shampoo is the first-line treatment.

Seborrheic dermatitis presents with yellow-white greasy scale, red and inflamed skin beneath, and a diffuse distribution extending from the scalp to the face — particularly the nasolabial folds, eyebrows, and ears. The itch has a burning quality. Body involvement may include the chest, back (in a pityriasis rosea-like pattern), and flexures. It responds strongly to antifungals and the first-line treatment is ketoconazole 2% shampoo combined with a topical steroid.

Scalp psoriasis presents with thick silver-white adherent scale, bright red sharply defined skin beneath, and sharp well-defined plaques that characteristically extend beyond the hairline onto the forehead and neck — this is the key distinguishing feature from dandruff. Itch is variable and can be severe. Body involvement includes elbows, knees, nails, and joints. It responds only partially to antifungals and requires betamethasone valerate scalp lotion combined with coal tar.

#### Step-By-Step Dandruff Treatment Protocol

Step 1 (Weeks 1–2): Use zinc pyrithione shampoo (e.g., Head and Shoulders Clinical) daily. Leave on for 3 minutes before rinsing. This alone clears 60–70% of mild dandruff.

Step 2 (Weeks 3–4): If dandruff persists, switch to ketoconazole 2% (Nizoral) shampoo twice weekly, using zinc pyrithione on other days. This combination addresses both fungal load and inflammation.

Step 3 (Weeks 5–8): If the scalp is red and irritated, add selenium sulfide 2.5% twice weekly. For severe seborrheic dermatitis, add betamethasone valerate 0.12% scalp application (Betnovate Scalp) for 2 weeks.

Step 4 — Maintenance (ongoing): Alternate between two shampoos (antifungal plus zinc pyrithione) weekly to prevent resistance. One treatment shampoo per week indefinitely prevents relapse in 85% of patients.

Step 5 — Natural adjuncts: Tea tree oil 5% shampoo has proven antifungal properties and is a useful adjunct. Aloe vera gel applied to the scalp soothes irritation. Coconut oil scalp masks should be used with caution — coconut oil may feed *Malassezia* and worsen symptoms in some patients.

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### 2.2 Hair Loss (Alopecia) — Comprehensive Guide

*Keywords: hair loss, alopecia, androgenetic alopecia, alopecia areata, telogen effluvium, minoxidil, finasteride*

**Expert Note (40 yrs):** For telogen effluvium in young women I always check serum ferritin — not just haemoglobin. Haemoglobin can be normal while ferritin is depleted. I treat aggressively if ferritin is below 70 mcg/L. Also: baricitinib for alopecia areata is the most exciting development I have seen in my career for this condition. After decades of having little to offer patients with totalis or universalis, we now have a treatment that genuinely regrows scalp and eyebrows.

Male androgenetic alopecia affects 50% of men by age 50. It is caused by genetic susceptibility and DHT sensitivity, which miniaturises follicles. The pattern follows the Hamilton-Norwood scale with bitemporal recession and crown thinning, with miniaturised hairs visible on trichoscopy. First-line treatment is oral finasteride 1 mg daily (a 5-alpha reductase inhibitor that reduces DHT by 70%), combined with topical minoxidil 5% once or twice daily. Second-line options include dutasteride 0.5 mg daily (more potent than finasteride) and low-level laser therapy. Hair transplant (FUE or FUT) is the definitive option. Hair loss is only partially reversible and is progressive without treatment.

Female androgenetic alopecia affects 40% of women by age 70. Causes include genetic predisposition and hormonal influences; PCOS and hyperandrogenism in younger women are important factors. The pattern follows the Ludwig scale with diffuse thinning at the crown and parting, with the frontal hairline preserved. First-line treatments are topical minoxidil 2–5% and spironolactone 100–200 mg/day (anti-androgen). Combined oral contraceptives, especially those containing drospirenone, are also beneficial. Second-line options include low-dose finasteride (off-label, 1–2.5 mg) and platelet-rich plasma (PRP) injections. Treatment stops progression but does not fully reverse it.

Alopecia areata (AA) has a lifetime prevalence of 2% and is caused by autoimmune CD8+ T-cell attack on the follicular bulb — it is HLA-associated. It presents as smooth, round bald patches with exclamation mark hairs at the margin; nail pitting is seen in 10–15% of cases. First-line treatment is intralesional triamcinolone acetonide 5–10 mg/mL (0.1 mL per cm<sup>2</sup> monthly) — the most effective option for limited disease. Baricitinib 4 mg once daily was FDA-approved in 2022 for severe AA, with 50% of patients achieving regrowth in trials. Ritlecitinib 50 mg and topical immunotherapy with DPCP are used for extensive disease. 80% of patients with limited AA regrow within a year, but 30% recur. Alopecia totalis and universalis have less than 10% spontaneous recovery.

Telogen effluvium is the most common cause of diffuse shedding. It is triggered by a physical or psychological stressor 2–3 months prior — illness, surgery, childbirth, crash dieting, thyroid disease, or iron deficiency. It presents with diffuse shedding of more than 100 hairs per day, a positive hair pull test, and mostly telogen (club-rooted) hairs. Treatment involves identifying and treating the trigger, iron supplementation if ferritin is below 70 mcg/L, nutritional support, and reassurance. Recovery typically occurs in 6–12 months once the trigger is resolved. If chronic (more than 6 months), investigate systematically for thyroid disease.

Traction alopecia is common in African and Caribbean women, the Sikh community, and ballet dancers due to chronic mechanical tension from tight braids, weaves, locs, and ponytails. It presents with frontal and temporal recession and a 'fringe sign' (short hairs at the hairline margin) with no inflammation in early stages. The first and most critical step is to stop all traction immediately. Topical minoxidil should be started early. Anti-inflammatory topical steroids can be used if perifollicular inflammation is present. Hair transplant is possible once scarring is established. Reversal is possible if caught early before scarring; permanent if treatment is delayed.

Scarring alopecias (including lichen planopilaris/LPP and central centrifugal cicatricial alopecia/CCCA) are uncommon but significantly under-diagnosed. LPP involves pain, itch, and burning with perifollicular erythema and scale; CCCA is associated with heat and traction combined with inflammation. Dermoscopy shows absence of follicular ostia (replaced by fibrosis) in affected areas. Treatment does not reverse

established scarring — the goal is to arrest progression. Hydroxychloroquine 200–400 mg once daily is first-line for LPP, combined with topical tacrolimus and intralesional steroids. Doxycycline 100 mg twice daily is anti-inflammatory. Ciclosporin or mycophenolate mofetil (MMF) is used for aggressive cases. JAK inhibitors are emerging as a promising new option.

## PART 3 — INFECTIOUS SKIN DISEASES

### 3.1 Bacterial Skin Infections

*Keywords: cellulitis, impetigo, folliculitis, MRSA, necrotizing fasciitis, bacterial skin infection, antibiotics*

Non-bullous impetigo is caused by *Staphylococcus aureus* and/or *Streptococcus pyogenes* and presents as honey-coloured crusted lesions with thin-walled vesicles that rupture to form crust on an erythematous base, most commonly on the face and limbs of children. Diagnosis is clinical, with a wound swab for culture if not responding to initial treatment. For mild-to-moderate cases, mupirocin 2% ointment (Bactroban) three times daily for 5–7 days is the topical gold standard. Severe cases require flucloxacillin 500 mg four times daily for 7 days; co-amoxiclav if polymicrobial; cefalexin as an alternative. Impetigo is highly contagious via direct contact and fomites; school exclusion is required until crusts have resolved.

Bullous impetigo is caused by *Staphylococcus aureus* phage type 71, which produces exfoliatin toxin. It presents as large, flaccid bullae that rupture to superficial erosions with less crusting than the non-bullous type; more common in neonates and infants. Treatment is flucloxacillin 500 mg four times daily for 7 days, with IV flucloxacillin for systemic spread; vancomycin or daptomycin IV for MRSA. Contagion risk is high.

Cellulitis is most commonly caused by Group A *Streptococcus* (*Streptococcus pyogenes*) and *Staphylococcus aureus*, accounting for 75% of cases. It presents as non-purulent spreading erythema, warmth, swelling, and pain with a poorly defined border, most commonly on the lower leg; fever occurs in severe cases. Mark the border with a skin pen to assess progression. Blood cultures if febrile; ultrasound to exclude abscess. For mild-to-moderate cases, amoxicillin-clavulanate or cefalexin for 5–7 days; clarithromycin if penicillin-allergic. Severe cases require IV benzylpenicillin plus flucloxacillin, or IV vancomycin if MRSA is suspected. Elevate the limb and manage predisposing factors such as oedema and tinea pedis. Cellulitis is not directly contagious person-to-person.

Erysipelas, caused by *Streptococcus pyogenes*, is distinguished from cellulitis by its sharply demarcated raised red plaque with an advancing, clearly defined red margin, accompanied by fever and rigors. It typically affects the face or lower leg. Oral phenoxymethylpenicillin 500 mg four times daily for 7–10 days; IV benzylpenicillin 2.4 g four times daily for severe cases.

Folliculitis is most commonly caused by *Staphylococcus aureus* (or *Pseudomonas aeruginosa* in hot-tub folliculitis). There is also a common fungal mimic — *Pityrosporum folliculitis* — which presents on the chest and back in gym users and must be treated with antifungals, not antibiotics. Diagnosis is clinical; culture and KOH testing if atypical or not responding. Topical mupirocin or fusidic acid plus chlorhexidine wash is the first-line approach; ketoconazole shampoo or wash daily for 2 weeks for *Pityrosporum folliculitis*. Oral flucloxacillin for bacterial cases; *Pseudomonas folliculitis* typically resolves spontaneously. Drain if a furuncle develops.

Furuncles, carbuncles, and abscesses are caused by *Staphylococcus aureus* (with MRSA increasingly common). A furuncle is a deep, painful nodule with a central pus point; a carbuncle is a cluster of furuncles, typically on the scalp, neck, buttocks, or thigh. Incision and drainage (I&D;) is the primary treatment. Culture pus and screen for MRSA in recurrent cases. For MRSA: TMP-SMX (Bactrim) DS twice daily for 5–7 days, doxycycline 100 mg twice daily, or clindamycin 300 mg three times daily if susceptible. For recurrent household MRSA, decolonise with mupirocin nasal ointment three times daily for 5 days combined with chlorhexidine body wash daily for 5 days.

## 3.2 Viral Skin Infections

*Keywords: herpes simplex, warts, HPV, molluscum, chickenpox, shingles, viral rash, antiviral treatment*

HSV-1 (oral herpes, cold sores) is transmitted via saliva and direct oral contact, with autoinoculation possible. It presents with a prodrome of tingling or burning followed by grouped vesicles on a red base at the vermilion border, crusting over 7–10 days. Starting antivirals at the first tingle — before the visible lesion appears — dramatically reduces severity and duration. For an acute single-day episode, valacyclovir 2 g twice daily for 1 day is highly effective if started in the prodrome. For suppression (more than 6 episodes per year), valacyclovir 500 mg once daily. SPF lip balm prevents UV-triggered recurrences.

HSV-2 (genital herpes) is transmitted sexually and vertically to neonates. It causes painful genital and perianal vesicles and erosions with dysuria and fever in primary infections; recurrent episodes are milder. Valacyclovir is the preferred antiviral. Primary episode: valacyclovir 1 g twice daily for 10 days. Recurrent episode: 500 mg twice daily for 3 days. Suppressive therapy at 500 mg once daily reduces transmission to partners by 50% — all patients with partners should know this. Disclose to sexual partners; condoms reduce but do not eliminate risk. Neonatal transmission risk is highest with primary infection in the third trimester.

Herpes zoster (shingles) is VZV reactivation, triggered by age over 60, immunosuppression, and stress. Treatment must begin within 72 hours of rash onset: valacyclovir 1 g three times daily for 7 days or famciclovir 500 mg three times daily for 7 days. If V1 dermatome (forehead or eye) is involved, refer to ophthalmology urgently. Pregabalin started early reduces post-herpetic neuralgia (PHN) risk — do not wait until PHN develops. The Shingrix recombinant zoster vaccine is 97% effective and is recommended for all patients over 50 in two doses 2–6 months apart.

Warts (HPV) are transmitted via skin contact, surfaces, and sexual contact. Common warts (HPV 2 and 4) present as rough papules on hands and fingers; plantar warts (HPV 1) are painful with thrombosed capillary dots; genital warts (HPV 6 and 11) form condylomata. HPV 16 and 18 are high-risk strains driving cervical, anal, and oropharyngeal cancers. There is no antiviral; clearance is immune-mediated and 80% of sexually active people will acquire HPV — most clear spontaneously. Topical salicylic acid 10–26% applied nightly for 12 weeks is first-line; cryotherapy every 3 weeks; imiquimod 5% cream for immune stimulation; podophyllin 10–25% applied by clinician for genital warts; laser or surgery for resistant cases. HPV vaccine (Gardasil-9) is nine-valent and protects against HPV 6, 11, 16, 18, 31, 33, 45, 52, and 58 — ideally given before sexual debut in three doses.

Molluscum contagiosum is transmitted via skin contact and shared towels. It presents as pearly, dome-shaped, umbilicated papules 1–5 mm in diameter. In children, lesions appear on the face, trunk, and limbs; in adults, mainly on the genitalia. In HIV/AIDS, lesions are widespread and large. In immunocompetent children, watchful waiting before aggressive treatment is recommended, as lesions clear spontaneously in 6–18 months. Exceptions are widespread cases causing distress or pre-school settings. Cantharidin ('beetle juice') applied by a clinician is highly effective; cryotherapy, curettage, imiquimod, and potassium hydroxide 5–10% solution are also used. Avoid shared towels and observe swimming pool precautions.

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## 3.3 Fungal Infections — Comprehensive Reference

*Keywords: ringworm, athlete's foot, tinea, candida, onychomycosis, antifungal, fungal nail, KOH scraping*

Tinea pedis (athlete's foot) is most commonly caused by *Trichophyton rubrum* (70%) and *Trichophyton interdigitale*. It presents in three patterns: interdigital (maceration and fissuring of the 4th–5th toe webs), moccasin type (diffuse scaling of the sole and sides of the foot), and vesicular (blistering on the instep). Diagnosis is clinical with KOH scraping of active edge; culture if uncertain. Topical terbinafine 1% cream

twice daily for 2 weeks or clotrimazole 1% cream twice daily for 4 weeks clears most cases. Oral terbinafine 250 mg once daily for 2 weeks is needed for the moccasin type or treatment failures. Always treat tinea pedis concurrently when treating onychomycosis — it is the number one cause of nail treatment relapse.

Tinea unguium (onychomycosis) is most commonly caused by *Trichophyton rubrum*. It presents with distal and lateral subungual yellow-white discolouration, subungual debris, thickening, and onycholysis. Diagnosis must be confirmed with culture or PCR before oral treatment (30% false-positive clinical diagnosis). Topical treatment is only suitable for superficial white or single nail disease using efinaconazole 10% solution or amorolfine 5% nail lacquer weekly for 6–12 months. Oral terbinafine 250 mg once daily is the most effective systemic option: 6 weeks for fingernails, 12 weeks for toenails. Itraconazole pulse therapy (200 mg twice daily for 1 week per month for 3 months) has fewer drug interactions. Monitor LFTs at baseline and 6 weeks. Set clear expectations: the nail will not look normal until the healthy nail grows out — 1 mm per month for toenails means full resolution takes up to 12 months.

Tinea corporis (ringworm) is caused by *Trichophyton rubrum* and *Microsporum canis* (from cats and dogs). It presents as circular or oval rings with a raised scaly advancing edge and central clearing; itchy, with multiple overlapping rings possible. KOH scraping from the active advancing edge confirms the diagnosis; culture if an animal source is suspected. Topical terbinafine cream twice daily for 1–2 weeks or clotrimazole twice daily for 4 weeks is usually sufficient. Oral terbinafine 250 mg for 2–4 weeks is needed for extensive or unresponsive disease; treat household animal sources. Critical warning: steroid plus antifungal combination creams (such as Canesten HC or Daktacort) used on tinea corporis cause Tinea Incognito — the inflammation is suppressed, masking the ring while the fungus spreads undetected. Never prescribe these for ringworm.

Tinea capitis (scalp ringworm) is caused by *Trichophyton tonsurans* (endothrix — common in North America and the UK) and *Microsporum canis* (ectothrix — common in Southern Europe and Africa). It presents as patchy alopecia with grey scale and a 'black dot' appearance from broken hairs at scalp level. Inflammatory kerion — a boggy mass — carries a scarring risk. Wood's lamp shows green fluorescence only for *Microsporum* species; it is negative for the more common *Trichophyton*. Fungal culture is the gold standard. Oral treatment is required — topical antifungals cannot penetrate the hair shaft. Use selenium sulfide shampoo to reduce spore shedding and prevent spread during treatment. Oral griseofulvin 10 mg/kg/day for 6–8 weeks, or terbinafine (less than 25 kg: 62.5 mg; 25–35 kg: 125 mg; over 35 kg: 250 mg daily) for 4–6 weeks. Screen and treat all household contacts — 30% of contacts are asymptomatic carriers who will reinfect the treated child.

Candidal intertrigo is caused by *Candida albicans* and presents as bright red macerated plaques in skin folds with satellite pustules (pathognomonic). It also causes genital candidiasis, nappy rash, and oral thrush (white plaques). KOH shows pseudohyphae and budding spores. Topical clotrimazole 1% or miconazole cream twice daily for 1–4 weeks plus nystatin cream or ointment; keep the area dry with barrier cream and absorbent powder. Oral fluconazole 150 mg single dose for vaginal candidiasis; weekly fluconazole 150 mg for 3–6 months for recurrent cases (more than 4 episodes per year). In recurrent vulvovaginal candidiasis, check HbA1c, exclude HIV, and review antibiotic and oral contraceptive use. Weekly fluconazole for 6 months reduces recurrence by 90%.

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### 3.4 Parasitic Skin Infections

*Keywords: scabies, head lice, leishmaniasis, hookworm, parasitic skin infection, permethrin, ivermectin*

Scabies is caused by the female *Sarcoptes scabiei* mite, which burrows into the stratum corneum to lay eggs. The hallmark is intense pruritus that is markedly worse at night — this nocturnal itch results from sensitisation

to mite faeces, which takes 4–6 weeks on first exposure. Dermoscopy shows the characteristic 'jet with contrail' sign at the burrow. Clinical features include linear burrows in finger webs, wrists, genitalia, nipples, and axillae — the face is spared in adults but affected in infants. Treatment is 5% permethrin cream applied to the entire body from the neck down, left on for 8–12 hours, and repeated on day 7. Oral ivermectin 200 mcg/kg in two doses 7 days apart is equivalent in efficacy with better compliance. All household members must be treated simultaneously regardless of symptoms. Wash all clothing, bedding, and towels at 60°C on day of treatment; bag non-washable items for 72 hours. Post-scabetic pruritus can persist for 4–6 weeks after successful treatment — this is an inflammatory reaction, not treatment failure. Prescribing a moderate topical steroid for the itch prevents many patients from being unnecessarily retreated.

Norwegian (crusted) scabies involves the same mite but in hyperinfestation (millions of mites versus the normal 10–15), occurring in immunocompromised or institutionalised patients. It presents as thick hyperkeratotic crusts on the hands, feet, scalp, and ears with minimal or no itch. Skin scraping shows masses of mites and eggs. This is highly contagious and can infect an entire nursing home or ward from a single case — always recognise early in elderly care home residents and do not assume 'dry skin.' Treatment is oral ivermectin 200 mcg/kg on days 1, 2, 8, 9, and 15, plus daily topical permethrin until crusts resolve, combined with keratolytics (urea 5–10%) to remove crusts. Strict contact precautions and environmental decontamination are required.

Head lice (*Pediculus capitis*) are caused by *Pediculus humanus capitis*. They cannot fly or jump. Pruritus of the scalp, neck, and behind the ears is the main symptom, with visible nits (white egg cases) glued to the hair shaft within 1 cm of the scalp that cannot be brushed off — distinguishing them from dandruff. Live lice confirm active infestation; nits alone indicate a previous infestation. First-line treatment in most countries is 4% dimeticone (Hedrin): apply for 8 hours or overnight, repeat in two doses 7 days apart — this works by physical suffocation with no possibility of resistance. 5% permethrin lotion (apply to dry hair, leave 10 minutes, rinse, and repeat on day 7) is an alternative, but permethrin resistance is widespread. Wet combing every 3 days for 2 weeks is an effective no-pesticide option. Treat all household members with live lice; no school exclusion is required in the UK (guidance changed in 2012). Check weekly for 2–3 weeks after treatment.

Cutaneous leishmaniasis is caused by *Leishmania* species (*L. major*, *L. tropica*, *L. mexicana*, *L. braziliensis*) transmitted by the *Phlebotomus* sandfly in tropical and subtropical zones. It presents as a painless papule at the bite site that enlarges with central ulceration and a raised, indurated 'volcano crater' rim, on the face and exposed limbs. Diagnosis requires slit-skin smear or biopsy for Giemsa staining showing amastigotes within macrophages; PCR is more sensitive. *L. major* heals spontaneously in 3–18 months, but *L. braziliensis* can cause destructive mucosal disease (espundia). Localised disease is treated with intralesional sodium stibogluconate (Pentostam) injections or topical paromomycin. Widespread, mucocutaneous, or immunocompromised disease requires liposomal amphotericin B (AmBisome) 3 mg/kg on days 1–5, 14, and 21 — the safest systemic option — or oral miltefosine 2.5 mg/kg/day for 28 days. It is not contagious person-to-person. Always consider leishmaniasis in returning travellers with non-healing ulcers from endemic regions including the Middle East, Central Asia, South America, and sub-Saharan Africa.

## PART 4 — SKIN CANCER & MELANOMA

### 4.1 Types of Skin Cancer — Expert Clinical Reference

*Keywords: skin cancer, melanoma, BCC, SCC, ABCDE rule, suspicious mole, Mohs surgery, immunotherapy skin cancer*

Skin cancer is the most common malignancy in the world. In my career I have excised over 15,000 skin cancer lesions. I have one unshakeable belief: most of these cancers were preventable and most deaths from melanoma were preventable with earlier diagnosis.

Basal cell carcinoma (BCC) accounts for approximately 75% of skin cancers with over 3 million cases per year in the USA. It originates from basal cells of the epidermis. It typically appears as a pearly or translucent nodule with a rolled telangiectatic border; central ulceration (the 'rodent ulcer') occurs in advanced cases. The morphoeic (scar-like) subtype is the most aggressive. Pigmented BCC can mimic melanoma. BCCs arise predominantly on the face (70%), particularly the nose, periorbital area, and ears, as well as the trunk. Metastasis is very rare (less than 0.1%), but local destruction can involve the orbit or cranium. Dermoscopy shows arborising telangiectasia, leaf-like structures, and spoke-wheel patterns. Diagnosis is by shave or punch biopsy.

Treatment for BCC: Mohs micrographic surgery is the gold standard for lesions in the H-zone of the face, recurrent tumours, and morphoeic BCCs. Standard excision with 4–5 mm margins is appropriate for low-risk nodular BCCs. Radiotherapy is an option for elderly or inoperable patients. Vismodegib or sonidegib (hedgehog pathway inhibitors) are used for advanced or metastatic BCC. Five-year survival is excellent at over 95%.

Squamous cell carcinoma (SCC) accounts for approximately 20% of skin cancers with over 1 million cases per year in the USA. It arises from squamous keratinocytes. It presents as a firm red nodule with a keratotic or scaly surface; ulceration occurs in advanced disease. Actinic keratosis is SCC in situ (Bowen's disease). SCCs arise predominantly on sun-exposed areas: the face, ears, dorsal hands, lips, and legs, as well as mucous membranes and on scars (Marjolin's ulcer). Metastasis risk is 2–5%, rising substantially on the lip (10–14%), ear, and in immunosuppressed patients (organ transplant patients face a 65-fold elevated risk). Dermoscopy shows glomerular and dotted vessels; biopsy is mandatory.

Treatment for SCC: Surgical excision with 4 mm margin for SCC in situ and 6 mm for low-risk tumours; Mohs surgery for high-risk or facial lesions. Radiotherapy is an option. Cetuximab is used for metastatic disease; cemiplimab (anti-PD-1) is FDA-approved for advanced and metastatic SCC. Five-year survival is over 90% for localised disease; 50–70% with nodal spread.

Melanoma accounts for only approximately 5% of skin cancers but is responsible for 75% of all skin cancer deaths. It arises from melanocytes. Clinically, it presents as an irregularly shaped pigmented lesion with ABCDE features. Nodular melanoma is often pink or red (amelanotic). Acral lentiginous melanoma occurs on the palms, soles, and subungual areas. Common sites are the back in men and the legs in women, but melanoma can also occur on the palm, nail, eye (ocular melanoma), anorectal mucosa, and mucosal surfaces. Metastasis risk increases dramatically with stage: Stage I 5%, Stage II 15–45%, Stage III 40–80% (lymph node involvement), Stage IV over 90% (distant metastases).

Diagnosis of melanoma requires dermoscopy followed by excisional biopsy with 2 mm margins — never a shave biopsy, as Breslow thickness must be assessed in full. Sentinel lymph node biopsy is required if Breslow thickness exceeds 0.8 mm. BRAF/NRAS/c-KIT mutation testing is essential for treatment planning. Treatment: Wide local excision with 1 cm margin for tumours less than 1 mm, and 2 cm margin for tumours

greater than 1 mm. Adjuvant immunotherapy with pembrolizumab or nivolumab (anti-PD-1) for Stage III. BRAF and MEK inhibitors (dabrafenib plus trametinib) for tumours with BRAF V600E mutation. Stage IV: combination immunotherapy with ipilimumab plus nivolumab. Five-year survival: Stage IA 97%, Stage IB 92%, Stage IIA 81%, Stage IIIA 78%, Stage IIIC 40%, Stage IV 20–28%.

**Expert Note (40 yrs):** *In 40 years I have diagnosed over 3,000 melanomas. The E — Evolving — is the one that saves lives. A 'boring' stable mole that suddenly changes is the most common story I hear. I tell every patient: 'I don't want you to panic about your moles — I want you to KNOW your moles.' Monthly self-examination under good light. Photograph them. A change you notice in 3 months could mean I catch a Stage I instead of Stage III. That difference is everything.*

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## 4.2 The ABCDE Rule — Melanoma Detection

*Keywords: ABCDE melanoma, mole check, asymmetry border colour diameter evolution*

A — Asymmetry: A normal mole has two halves that mirror each other. A melanoma warning sign is when one half is distinctly different from the other. Draw a mental line through the middle — the two halves should not differ significantly in size, shape, or colour.

B — Border: A normal mole has smooth, even, well-defined edges. The melanoma warning sign is an irregular, ragged, notched, blurred, or poorly defined border — the edge looks like the pieces of a jigsaw rather than a smooth circle.

C — Colour: A normal mole has a single shade of brown or tan. The melanoma warning sign is multiple colours within a single lesion — various shades of brown, black, and tan, plus areas of white, red, or blue. A mole that was once uniformly brown but now has a black spot in one corner and a reddish area in another warrants urgent review.

D — Diameter: A normal mole is usually smaller than 6 mm (the size of a pencil eraser). Any lesion larger than 6 mm that is new or growing is a warning sign. However, early melanomas can be smaller — size alone is insufficient as a criterion. Nodular melanoma may be less than 5 mm and still lethal.

E — Evolving: A normal mole is stable in appearance over months and years. ANY change in size, shape, or colour, or ANY new symptoms (bleeding, itching, crusting, tenderness) over weeks or months is the most important warning sign. 'This mole has been there for years but it started itching 3 months ago and seems darker' — this E criterion alone warrants urgent referral, regardless of the other ABCDE features.

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## 4.3 Staging, Treatment & Prevention

*Keywords: melanoma staging, skin cancer prevention, sun protection, SPF, skin cancer screening*

Prevention of skin cancer is the most impactful intervention available. UV radiation is the primary driver of BCC, SCC, and melanoma. SPF 50+ broad-spectrum sunscreen applied every morning (1/4 teaspoon for the face and neck) and reapplied every 2 hours outdoors is the single most evidence-based anti-ageing and cancer prevention step available. 90% of visible facial ageing — wrinkles, pigmentation, and collagen loss — is caused by UV, not by chronological ageing.

Additional sun protection measures include seeking shade during peak UV hours (10 am to 4 pm), wearing protective clothing (long sleeves, wide-brimmed hats), wearing UV-blocking sunglasses to prevent ocular melanoma, and avoiding sunbeds — which increase melanoma risk by 75% when first used before age 35. Annual full-body skin examinations are recommended for high-risk individuals: those with a personal or family

history of melanoma, more than 50 moles, history of significant childhood sunburn, or immunosuppression.

## PART 5 — ALLERGIC & DRUG REACTIONS

### 5.1 Urticaria & Angioedema

*Keywords: hives, urticaria, angioedema, allergic rash, antihistamine, omalizumab, anaphylaxis*

Acute spontaneous urticaria (less than 6 weeks) presents with individual wheals lasting less than 24 hours recurring in crops. Common triggers include foods (nuts, shellfish, eggs), drugs (penicillin, NSAIDs, aspirin), viral infections, insect stings, and idiopathic causes (50%). First-line treatment is a non-sedating antihistamine taken daily (not as needed): cetirizine 10 mg, loratadine 10 mg, or fexofenadine 180 mg. If not controlled, the antihistamine dose can be increased up to 4 times the licensed dose (off-label but guideline-supported). A short course of prednisolone 40 mg for 3–5 days is used for severe episodes. For anaphylaxis: intramuscular epinephrine 0.5 mg (1:1000), call 999/911, IV hydrocortisone, and chlorphenamine IV or IM.

Chronic spontaneous urticaria (more than 6 weeks, occurring daily or near-daily) is caused by autoimmune mechanisms (IgE or IgG anti-FcεRI) in 40% of cases, with associations with thyroid disease, H. pylori, and NSAID intolerance; the remainder are idiopathic. First-line management is maximising the non-sedating antihistamine to 4 times the dose (e.g., cetirizine 40 mg daily). If uncontrolled, omalizumab (Xolair) 300 mg subcutaneously every 4 weeks achieves 65–70% complete response. Ciclosporin 3–5 mg/kg is used for truly refractory cases; dapsone for pressure urticaria.

Urticarial vasculitis differs from standard urticaria in that wheals persist for more than 24 hours and leave bruising. It is caused by SLE, hepatitis C, medications, or IgM/IgG immune complexes; hypocomplementaemic urticarial vasculitis is the most serious form. Treat with oral colchicine or hydroxychloroquine; NSAIDs for mild disease. Dapsone, systemic steroids, and rituximab are used for hypocomplementaemic urticarial vasculitis.

Cold urticaria is triggered by cold contact or cold air. It is idiopathic in most cases, with cryoglobulinaemia and secondary cold agglutinins as identifiable causes. Critically, drowning risk from systemic cold exposure (cold water swimming) is a genuine life-threatening risk that must be communicated clearly. First-line treatment is fexofenadine 180 mg daily and strict cold avoidance. Cyproheptadine, rupatadine (which has platelet-activating factor antagonism), and omalizumab are second-line options. All patients should carry an epinephrine auto-injector.

**SJS/TEN/DRESS are dermatological emergencies. Mortality: SJS 5–10%; TEN 25–35%; DRESS 10%. The first action in ALL cases is to STOP the causative drug immediately. Every hour of delay worsens outcome.**

### 5.2 Serious Drug-Induced Skin Reactions — Recognition & Emergency Management

*Keywords: Stevens-Johnson syndrome, SJS, TEN, DRESS, drug rash, drug allergy, skin peeling, drug reaction*

Morbilliform drug eruption (maculopapular measles-like rash) is the most common drug reaction. It presents as mild itching starting on the trunk with no mucosal involvement and a generally well patient. Common culprit drugs include amoxicillin/ampicillin (in 90% of cases given with EBV mononucleosis), allopurinol, antiepileptics, and sulfonamides. Management: stop the drug if severe or non-essential; cetirizine 10 mg;

moderate topical steroid; usually self-resolves in 1–2 weeks. Mortality is less than 1%.

Fixed drug eruption presents as single or few round hyperpigmented macules or plaques that reappear at exactly the same site on re-challenge. Mucosa may be involved. Common culprit drugs include NSAIDs (ibuprofen, naproxen), tetracyclines, paracetamol, phenolphthalein, and trimethoprim. Management: permanently identify and avoid the causative drug; cool compress; moderate-potency topical steroid for acute flare. Mortality is negligible.

Stevens-Johnson Syndrome (SJS) involves painful skin blistering with mucosal erosions (oral, ocular, and genital) with less than 10% BSA skin detachment. Systemic illness and fever are prominent. The Nikolsky sign is positive. Target lesions with vesicular or necrotic centres are characteristic. The most common culprit drugs globally are allopurinol (number one cause), lamotrigine, carbamazepine, phenytoin, sulfonamides, nevirapine, and NSAIDs of the oxicam class. Emergency management: stop the drug immediately, admit to hospital, seek urgent ophthalmology review (ocular scarring is a major complication). IV fluids and nutritional support. Ciclosporin 3 mg/kg/day or IVIG 1 g/kg/day for 3 days. Avoid systemic steroids (controversial — may increase sepsis risk, particularly in TEN). Mortality 5–10%.

Toxic Epidermal Necrolysis (TEN) is the same process as SJS but with more than 30% BSA detachment. Sheets of skin detach, high fever is present, and multi-organ involvement occurs, resembling full-thickness burns. Causative drugs are the same as SJS; allopurinol is most common in Asian populations carrying the HLA-B\*5801 allele. Mandatory transfer to ICU or Burns Unit. The SCORTEN score guides prognosis (1 point each for: age over 40, heart rate over 120, cancer, BSA involvement over 10%, BUN over 10, bicarbonate below 20, glucose over 14). Treatment: IVIG 2–3 g/kg over 3 days; ciclosporin, etanercept — evidence is growing. Meticulous moist wound care. Eye care every 2 hours. Mortality 25–35%.

DRESS (Drug Reaction with Eosinophilia and Systemic Symptoms) presents as a widespread morbilliform rash combined with fever, lymphadenopathy, and one or more organ involvements — the liver is most common, followed by kidneys, lungs, and heart. It is characterised by eosinophilia and atypical lymphocytes, with slow onset (typically 2–6 weeks after drug initiation). Common culprit drugs include carbamazepine, lamotrigine, phenytoin, allopurinol, vancomycin, minocycline, abacavir, and efavirenz. Emergency management: stop the drug immediately. Start systemic corticosteroids — prednisolone 1–2 mg/kg/day — and taper slowly over 8–12 weeks (rapid taper causes relapse). Monitor LFTs, FBC, renal function, and troponin weekly initially. DRESS can reactivate HHV-6, CMV, and EBV. Mortality is approximately 10%, higher with myocarditis or fulminant hepatic failure.

## PART 6 — LIFE-THREATENING SKIN CONDITIONS

**NECROTIZING FASCIITIS IS A SURGICAL EMERGENCY. Every hour without debridement increases mortality by 3–9%. If this diagnosis is considered — call the surgical team NOW. Do not wait for imaging. Do not wait for labs.**

### 6.1 Necrotizing Fasciitis — The Flesh-Eating Disease

*Keywords: necrotizing fasciitis, flesh eating disease, skin emergency, rapidly spreading infection*

In 40 years I have been called to the ward or A&E; for suspected necrotizing fasciitis (NF) approximately 80 times. The hardest lesson I learned early in my career: do not let the skin appearance reassure you. The hallmark of NF is that the PAIN is vastly out of proportion to the external skin findings — especially in the early stages. By the time the skin looks dramatically necrotic, the fascia underneath is already destroyed for centimetres beyond the visible margin.

Early stage (6–24 hours): The skin shows erythema, warmth, and swelling that is indistinguishable from severe cellulitis; the skin is intact. The critical feature is SEVERE DISPROPORTIONATE PAIN. Systemic signs include fever above 38°C, tachycardia, and elevated WCC, CRP, and lactate. A LRINEC score of 6 or more strongly suggests NF. Action: surgical consultation immediately, IV broad-spectrum antibiotics, blood cultures. CT scan if diagnosis is uncertain — but do not delay surgery for imaging.

Intermediate stage (24–72 hours): Blistering and haemorrhagic bullae appear. Crepitus on palpation indicates subcutaneous gas. The skin develops a bronze or grey discolouration. Loss of sensory nerve function means the area paradoxically becomes less painful — do not be falsely reassured by this. Systemic signs include septic shock, multi-organ dysfunction, DIC, and severely elevated inflammatory markers. Action: emergency surgical debridement removing ALL necrotic tissue; re-explore every 24–48 hours; ICU care; IV piperacillin-tazobactam or meropenem combined with clindamycin (to inhibit toxin production) and vancomycin if MRSA.

Late stage (over 72 hours): Frank skin necrosis, black eschar, and extensive tissue destruction with gaping wounds. Shock, organ failure, DIC, and death if not treated. Continue aggressive debridement; reconstruct with skin grafts after infection is controlled; hyperbaric oxygen as an adjunct. Mortality is 30–40% even with treatment.

### 6.2 Stevens-Johnson Syndrome & Toxic Epidermal Necrolysis

These conditions are covered in detail in Part 5.2. SJS involves less than 10% BSA skin detachment (mortality 5–10%); TEN involves more than 30% BSA detachment (mortality 25–35%). Both require immediate drug cessation, hospital admission, and ophthalmology referral. TEN mandates ICU or Burns Unit transfer. Ciclosporin and IVIG are the evidence-based systemic treatments.

### 6.3 Toxic Shock Syndrome

Toxic shock syndrome (TSS) is mediated by staphylococcal (TSST-1) or streptococcal (SpeA) superantigens causing massive cytokine release and multi-organ failure. Staphylococcal TSS mortality is 3–5%; streptococcal TSS mortality can reach 50%. Cardinal signs include fever of 38.9°C or higher, diffuse macular

erythroderma (a 'sunburn rash'), and hypotension. Desquamation of the palms and soles occurs 1–2 weeks later. Emergency management: remove the source (tampon or foreign body), IV flucloxacillin plus clindamycin, IVIG 1–2 g/kg, ICU care, and vasopressors for shock.

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## 6.4 Pemphigus Vulgaris

Pemphigus vulgaris is caused by IgG autoantibodies to desmoglein, producing full-thickness epidermal acantholysis. Mortality is less than 5% with modern treatment but was historically 75%. Cardinal signs are flaccid blisters, painful oral erosions preceding skin involvement, and a positive Nikolsky sign. Direct immunofluorescence (DIF) shows IgG deposits at the dermal-epidermal junction. Emergency treatment for severe disease: high-dose prednisolone 1 mg/kg combined with rituximab 1000 mg IV given twice 2 weeks apart. IVIG is used for life-threatening presentations. Meticulous wound care and oral hygiene are essential.

Meningococcal septicaemia (purpuric rash) is caused by *Neisseria meningitidis* septicaemia with DIC. Mortality is 10% for meningococcaemia but up to 70% in fulminant purpura fulminans. The cardinal sign is a non-blanching purpuric or petechial rash with fever, neck stiffness, and photophobia. The 'glass test' — the rash does NOT blanch under glass — is the classic emergency diagnostic criterion. Give IV or IM benzylpenicillin immediately, even before hospital. Cefotaxime or ceftriaxone IV; ICU; dexamethasone 4 mg four times daily for 4 days. MenACWY and MenB vaccination for household contacts.

Staphylococcal scalded skin syndrome (SSSS) is caused by staphylococcal exfoliatin toxins A and B, which cleave desmoglein 1 in the granular layer, causing a superficial split. Mortality is less than 5% in children but up to 60% in adults. Key features are fever, tender skin, perioral fissuring, and superficial peeling. The Nikolsky sign is positive but, critically, there are no mucosal erosions — this is the key feature distinguishing SSSS from TEN. Treatment: IV anti-staphylococcal antibiotics (flucloxacillin or vancomycin for MRSA), fluid and electrolyte support, and wound care.

## PART 7 — SKIN HEALTH, CARE & PRACTICAL SOLUTIONS

### 7.1 Skin Types & Evidence-Based Care Protocols

*Keywords: skin type, oily skin, dry skin, combination skin, sensitive skin, skincare routine, skin barrier*

Normal skin is characterised by balanced sebum, fine pores, few blemishes, good elasticity, and a comfortable feel. It may develop pigmentation or photoageing over time. Cleanse with a gentle gel or cream cleanser at pH 5.5; moisturise with a light lotion or cream containing ceramides and niacinamide; apply SPF 30+ broad-spectrum sunscreen every morning. Active ingredients to use are vitamin C serum in the morning and retinol (2–3 nights per week) in the evening. Avoid over-cleansing, harsh scrubs, and skipping SPF.

Dry skin presents as dull, tight, rough, and flaky, with fine lines exaggerated and sensitivity to cold and wind. It is prone to eczema. Use a gentle creamy non-foaming cleanser free from SLS as a soap substitute; apply a rich cream or ointment containing ceramides, hyaluronic acid, and shea butter to damp skin for maximum penetration. Apply SPF 30+ in a hydrating base or separately. Active ingredients include hyaluronic acid serum, glycerol, ceramide repair creams, and omega-3 fatty acid supplementation. Avoid hot water showers, alcohol-containing products, fragrance, and retinol without adequate moisturisation.

Oily skin presents as a shiny T-zone, enlarged pores, blackheads, and a tendency to acne and folliculitis. Oily skin actually ages better than dry skin because oil provides natural moisture. Use a gel or foaming cleanser with salicylic acid 0.5–2% and niacinamide 4% at pH 5.5, twice daily. Moisturise with an oil-free, non-comedogenic gel moisturiser with zinc niacinamide for sebum control. Apply a mattifying SPF 30+ in a water-based or gel formula, reapplying with matte powder midday. Salicylic acid BHA toner (0.5–2%) and niacinamide 10% are the key active ingredients. Avoid heavy oils (coconut, almond), petrolatum on the face, and over-washing, which strips oil and triggers rebound sebum production.

Combination skin — the most common skin type, affecting approximately 50% of the population — presents as an oily T-zone with normal or dry cheeks, with prominent seasonal changes. Use a gentle balanced gel cleanser for the whole face; apply zone-specific moisturisers (gel or mattifying product on the T-zone, cream on the cheeks). A light SPF 30+ fluid or light lotion works for the whole face; mattifying powder on the T-zone during the day. Use salicylic acid targeted on the T-zone only and niacinamide across the whole face; retinol on the T-zone with barrier repair focused on the cheeks. Avoid applying the same heavy product to both zones and do not skip moisturiser on the T-zone, which causes compensatory oil production.

Sensitive skin is easily irritated with stinging, burning, and redness after new products. It has a thin and fragile barrier with a history of allergy or reactivity, and is associated with rosacea, contact dermatitis, perioral dermatitis, and couperose skin. Use an ultra-gentle, fragrance-free, hypoallergenic cream cleanser or micellar water as an alternative; cleanse only once per day. Moisturise with a minimal-ingredient, fragrance-free barrier repair cream containing ceramide NP, cholesterol, and fatty acids (La Roche-Posay Toleriane, CeraVe, or Aveeno Eczema). Use mineral SPF (zinc oxide or titanium dioxide) — less irritating than chemical UV filters; tinted formulas reduce the need for makeup. Active ingredients to introduce slowly (after barrier is repaired) include colloidal oatmeal, centella asiatica (cica), panthenol, and azelaic acid 10%. Avoid fragrance, alcohol, essential oils, AHAs and BHAs until the barrier is repaired, and exfoliating brushes.

### 7.2 Daily Skincare Routine — Expert Protocol

*Keywords: skincare routine, morning skincare, evening skincare, retinol, vitamin C, SPF routine, anti-aging*

## Morning Routine

Step 1 — Cleanse: Use a gentle pH-balanced cleanser (pH 5–5.5) without SLS or fragrance. Cleanse for 30–60 seconds with lukewarm water and pat dry (never rub). This removes overnight sebum, microbes, and pillow bacteria.

Step 2 — Vitamin C Serum (optional but powerful): Apply L-ascorbic acid 10–20%, or stabilised forms such as sodium ascorbyl phosphate or ascorbyl glucoside. The most effective and stable combination is vitamin C with vitamin E and ferulic acid. Apply 3–5 drops on dry skin and wait 60 seconds to absorb; use only air-tight packaging and refrigerate if possible. Vitamin C acts synergistically with SPF by neutralising UV-generated free radicals that SPF cannot intercept.

Step 3 — Moisturise: Apply a moisturiser appropriate for your skin type (see section 7.1), containing ceramides, niacinamide, hyaluronic acid, and glycerin. Apply with fingertips in upward strokes, including the neck, 1–2 minutes after serum. This locks in the serum and provides a base for SPF application.

Step 4 — Eye cream (optional): Use peptides, caffeine, vitamin K, or low-concentration retinol 0.025%. Must be fragrance-free and ophthalmologist-tested. Apply with the ring finger (the gentlest) to the orbital bone — not on the eyelid itself. Morning eye cream with caffeine reduces puffiness. The periorbital skin is the thinnest on the face and shows ageing earliest.

Step 5 — Sunscreen (MANDATORY — the single most evidence-based anti-ageing step): Use SPF 50+ broad-spectrum UVA+UVB. Mineral filters (zinc oxide or titanium dioxide) or chemical filters (avobenzone or tinosorb) with a PA++++ UVA rating. Apply 1/4 teaspoon (1.25 mL) for the face and neck. Apply 15 minutes before sun exposure and reapply every 2 hours outdoors. Use every single morning even indoors — UVA penetrates glass. 90% of visible facial ageing is caused by UV radiation, not chronological ageing.

## Evening Routine

Step 1 — Double cleanse (if you wore SPF or makeup): First, use micellar water or a cleansing oil or balm on dry skin to emulsify SPF and makeup; add water, massage, and rinse. Then follow with a regular cleanser. SPF is oil-based — water-based cleansers alone remove less than 60% of SPF. Double cleansing removes 99% or more.

Step 2 — Chemical exfoliant (2–3 nights per week, not every night): AHAs such as glycolic acid (5–10%) and lactic acid (5–12%) are ideal for brightening, fine lines, and dry skin. BHAs such as salicylic acid (0.5–2%) penetrate pores and are ideal for acne-prone and oily skin. PHAs such as gluconolactone are the gentlest and are suitable for sensitive skin. Apply after cleansing; leave-on formulas for 5–10 minutes. Do not combine with retinol on the same night. Signs of over-exfoliation are burning, shiny tight skin, and increased sensitivity — stop all actives for 1–2 weeks and repair the barrier.

Step 3 — Retinoid (the number one proven anti-ageing ingredient for photoageing): OTC options include retinol 0.25–0.5% (building to 1%), retinaldehyde 0.05–0.1% (more potent with less irritation), and retinyl esters. Prescription options include tretinoin 0.025–0.1% (the most evidenced) and adapalene 0.3%. Start with 1 night per week for 2 weeks, then 2 nights per week, building to nightly over 3–6 months. For sensitive skin, use the 'sandwich' method: moisturiser, then retinoid, then moisturiser. Avoid in pregnancy. Retinoids cause initial purging (acne worsens in weeks 1–6) and peeling — this is not a reason to stop. 90% of patients who persist past the irritation phase see transformative results by month 3–4.

Step 4 — Night moisturise: Use a night cream, sleeping mask, or regular cream with ceramides, peptides (matrixyl, argireline), hyaluronic acid, niacinamide, and squalane. Apply as the last step after retinoid, sealing in all actives. Focus on the neck and décolletage. The PM moisturiser seals the skin during the window of peak cell renewal (midnight to 4 am) — use your most nutritive product at night.

Step 5 — Targeted treatments (as needed): Benzoyl peroxide 2.5–5% spot treatment on individual acne lesions only — never sleep with benzoyl peroxide on the whole face as it bleaches bedding. Azelaic acid 20% for PIH, rosacea, and acne. Clay mask for 10 minutes twice weekly for oily skin. Face oil (squalane, rosehip) for dry skin types.

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## 7.3 Lifestyle & Nutrition for Optimal Skin Health

*Keywords: diet skin health, hydration skin, sleep skin, stress skin, smoking skin, omega-3 skin*

**Hydration:** Dehydration causes dull, flaky, fine-line-accentuated skin and impairs wound healing and barrier function. Reduced interstitial fluid decreases dermal turgor and impairs keratinocyte function. Drink 1.5–2.5 L of water per day, adjusting for exercise and climate. Limit caffeine and alcohol. Eat water-rich foods such as cucumber, melon, and leafy greens. Monitor urine colour — pale yellow indicates good hydration.

**Omega-3 fatty acids** reduce inflammatory dermatoses including eczema, psoriasis, and acne by strengthening the lipid bilayer of the skin barrier and reducing trans-epidermal water loss (TEWL). They compete with arachidonic acid and reduce leukotriene B4 and prostaglandin E2 production — key inflammatory mediators in skin disease. Eat fatty fish (salmon, mackerel, sardines) 2–3 times per week, or supplement with fish oil 1–3 g of EPA+DHA daily. Flaxseed and chia seed provide ALA (plant omega-3), which is less bioavailable than marine omega-3.

**Vitamin C** is essential for collagen synthesis (hydroxylation of proline and lysine), acts as an antioxidant, aids wound healing, and reduces pigmentation. It is a cofactor for prolyl and lysyl hydroxylase enzymes and reduces melanin oxidation. As the plasma half-life is short, daily intake from citrus fruits, kiwi, strawberries, bell peppers, and broccoli is required. Topical vitamin C serum acts directly on the skin in addition to dietary sources.

**Zinc** regulates sebum production, has anti-inflammatory properties, and is critical for wound healing. It inhibits 5-alpha-reductase (similar mechanism to finasteride) and modulates TLR2 inflammation in acne. Good dietary sources include pumpkin seeds, oysters, beef, legumes, and nuts. Supplement with zinc gluconate 30 mg if deficiency is suspected. Zinc deficiency causes hair loss (telogen effluvium), poor wound healing, and perioral dermatitis.

**High glycaemic index diet** worsens acne and accelerates photoageing through advanced glycation end-product (AGE) formation and collagen crosslinking. Glucose spikes trigger insulin and IGF-1 elevations, driving androgen production and sebum increase. AGEs crosslink collagen fibres, making them stiff and yellow. Reduce refined carbohydrates, sugar, white bread, and white rice; choose low-glycaemic alternatives such as wholegrains, legumes, and vegetables. The Mediterranean diet pattern has the most anti-inflammatory evidence in dermatology.

**Sleep:** Sleep deprivation impairs barrier recovery, worsens eczema and psoriasis, increases cortisol (which drives acne), and reduces growth hormone secretion needed for cell repair. Cortisol increases sebum and acne; reduced growth hormone means slower collagen synthesis; elevated TNF-alpha triggers inflammatory dermatosis flares. Aim for 7–9 hours per night. A silk or satin pillowcase reduces friction and moisture absorption compared with cotton. Change pillowcase twice weekly (bacteria accumulate rapidly). Sleep on your back if possible to minimise mechanical facial pressure.

**Stress:** Cortisol and substance P trigger mast cell activation, leading to itch, urticaria, eczema flares, psoriasis flares, and acne. Stress impairs skin immune surveillance through HPA axis activation; CRH directly stimulates sebocytes; neuropeptides (substance P, CGRP) trigger mast cells. Stress delays wound healing by up to 40%. Mindfulness-based stress reduction is proven in RCTs to reduce psoriasis flares.

Regular aerobic exercise, adequate sleep, psychological support for chronic dermatoses, and identifying personal trigger patterns are all important interventions.

**Smoking:** Nicotine causes vasoconstriction that reduces skin oxygen by 50%; free radicals destroy elastin and collagen; cadmium in smoke inhibits superoxide dismutase; and matrix metalloproteinases (MMPs) are upregulated. The clinical result is a sallow, grey complexion, premature perioral wrinkles, dramatic worsening of psoriasis, increased squamous cell carcinoma risk, delayed wound healing, and Buerger's disease (digital ischaemia). Cessation is the single most impactful lifestyle intervention for skin health in smokers — skin begins improving within 4 weeks of stopping.

**Alcohol:** Vasodilation causes flushing and worsens all vascular skin conditions. The diuretic effect causes dehydration; antioxidants are depleted; oxidative stress increases; and sleep quality is disrupted with secondary skin effects. Alcohol is the number one rosacea trigger in my clinic. Limit to 14 units per week; red wine and beer are worse than spirits for rosacea. Hydrate between drinks and avoid alcohol after intense sun exposure.

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## 7.4 Home Remedies — Evidence-Based Assessment from 40 Years of Clinical Practice

*Keywords: aloe vera skin, coconut oil skin, tea tree oil, turmeric skin, home remedies dermatology, natural skincare*

Over 40 years, thousands of patients have asked me about natural remedies. My assessment: some have real scientific merit as adjuncts; many are harmless but ineffective; a few are actively harmful. The fundamental principle I apply is: what is the evidence, and what is the risk? Natural does not mean safe — poison ivy is natural.

**Aloe vera gel (evidence level ★★★★★):** Fresh or stabilised aloe vera gel has genuine evidence as an adjunct for sunburn — the acemannan polysaccharide has anti-inflammatory and humectant properties. For minor burns, it reduces pain and promotes healing. For eczema, it is soothing as an adjunct but does not replace medical treatment. Caution: the yellow sap (aloe latex) from under the skin can cause contact dermatitis and is toxic if ingested; stabilised gel can cause allergy in some patients. Verdict: Use for sunburn and minor irritation. Good adjunct, not a substitute for eczema treatment.

**Coconut oil (evidence level ★★★ for moisturising):** Coconut oil is a good occlusive moisturiser that reduces trans-epidermal water loss and is safe as a body moisturiser, including in eczema patients who cannot afford other emollients. Lauric acid has mild antimicrobial properties. However, it is comedogenic on the face (clogs pores and causes acne) and feeds *Malassezia* yeast — it worsens pityriasis versicolor and seborrheic dermatitis. Verdict: Good body moisturiser for eczema. Avoid on the face and scalp.

**Tea tree oil (evidence level ★★★★★):** At 5% concentration, tea tree gel is equivalent to 5% benzoyl peroxide for acne in one RCT, with slower onset but fewer side effects. A 5% shampoo reduces *Malassezia* and is effective for dandruff. A 25–50% solution is active against *Trichophyton rubrum* for athlete's foot. Must always be diluted — neat tea tree oil causes contact dermatitis and chemical burns. Toxic if ingested, especially in children. Verdict: Good adjunct for mild acne and dandruff at 5% concentration. Never use neat on skin.

**Turmeric/curcumin (evidence level ★★):** Good preclinical (lab and animal) anti-inflammatory evidence, with some pilot studies in psoriasis and acne. Oral curcumin shows mild benefit for psoriasis in small trials. However, bioavailability is poor (piperine or fat improves absorption); it stains skin yellow; no large-scale RCT data exist; and topical application leaves an orange stain. Verdict: Interesting future potential but insufficient evidence for clinical recommendation. Not harmful as a dietary spice.

Apple cider vinegar (evidence level ★): I actively discourage the use of ACV on skin. The evidence is very limited — the acidic nature (pH 2.5) might theoretically help restore skin pH 5.5, but regular use disrupts the skin barrier. I have seen at least 50 patients over my career with ACV-induced chemical burns on skin from undiluted application. There is no RCT evidence for any skin condition. Verdict: Do not use. The chemical burn risk is real and the evidence does not support any skin benefit.

Colloidal oatmeal (evidence level ★★★★★): One of the best-evidenced natural skin ingredients in dermatology. FDA-approved as a skin protectant. Proven anti-inflammatory properties from avenanthramides; protease inhibitor properties reduce the itch cascade; forms a protective film on skin. Widely used in dermatology for 2,000 years. Oat allergy is rare — patients with coeliac or gluten sensitivity can usually tolerate topical application as the proteins differ from gluten. Verdict: Highly recommend. Add to bath water for eczema or chickenpox; look for colloidal oatmeal in moisturisers (Aveeno, CeraVe).

Medical-grade Manuka honey (evidence level ★★★★★): Medical-grade Manuka honey (UMF 15+/20+) has proven antimicrobial activity against MRSA, reduces wound biofilm, and accelerates healing in chronic wounds. It is approved in EU and UK wound care guidelines. Standard supermarket honey has variable and lower activity and is not equivalent. Must be medical grade for wound care; not suitable for deep wounds without clinical supervision; contraindicated in honey allergy. Verdict: Legitimate medical wound care agent. I use it in my clinic for wound care in penicillin-allergic patients with infected eczema.

Ice and cold compress (evidence level ★★★): Vasoconstriction reduces acute inflammation; cold competes with itch signals at C-fibres; helpful for acute urticaria and reduces post-traumatic swelling. Never apply ice directly to skin — cryoinjury risk is real (the same process used therapeutically in cryotherapy to destroy tissue). Always wrap in cloth. No therapeutic benefit beyond acute symptom relief. Verdict: Safe and effective for immediate itch and swelling relief. Good first aid for acute flares while awaiting medication.

Zinc oxide paste (evidence level ★★★★★): Physical UV blocker; anti-inflammatory; mild astringent; excellent barrier protection; antimicrobial. The first-choice treatment for nappy (diaper) rash. Widely used in wound care for peristomal care and skin protection in incontinent patients. Zinc oxide SPF is the first choice for sensitive skin and rosacea due to minimal irritation. Verdict: I routinely recommend zinc oxide paste for nappy rash and skin protection. Zinc oxide SPF is my first choice for patients with sensitive or rosacea-prone skin.

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## 7.5 When to See a Dermatologist — Clinical Urgency Guide

*Keywords: when to see dermatologist, urgent referral, skin emergency, self-care, mole check, skin cancer screening*

**Expert Note (40 yrs):** After 40 years the advice I give to every patient leaving my clinic is the same: 'Know your skin. Love your skin. Protect your skin. And when something changes — come back.' Early diagnosis is not just better for outcomes — in melanoma it is the difference between life and death. I have never met a patient who wished they had waited longer to get a mole checked. The sun you enjoy at 25 presents its bill at 55. SPF 50 every single day is the best investment in your skin you will ever make. I have been practising dermatology for four decades and I still apply it every morning before I leave the house.

**EMERGENCY — Call 999/ER Now:** Rapidly spreading red or swollen skin with fever; skin peeling off in sheets; throat swelling or difficulty breathing; extreme disproportionate pain in a skin area; non-blanching purple or red rash with fever. Conditions to consider: Necrotizing fasciitis, TEN/SJS, Anaphylaxis, Meningococcal septicaemia, Pemphigus flare. Call emergency services immediately. Do not drive yourself.

**URGENT** — Same Day / A&E; (within 24 hours): A painful blistering rash in a single stripe (possible shingles — antivirals must begin within 72 hours, this window is critical); a rapidly growing skin lump; a widespread allergic reaction with hives plus facial swelling; fever with an infected skin area; eye involvement with a skin condition. Conditions to consider: Herpes zoster, Angioedema, Orbital cellulitis, early SJS/TEN, Eczema herpeticum. See your GP on the same day or go to A&E; if the GP is unavailable.

**SOON** — 1 to 2 weeks: Any mole showing ABCDE changes; a non-healing sore or ulcer lasting more than 4 weeks; a new or changing skin growth; severe or scarring acne unresponsive to over-the-counter treatment; hair loss with scalp symptoms; psoriasis that is inadequately controlled. Conditions to consider: Melanoma/SCC/BCC, Actinic keratosis, Scarring alopecia, Moderate-severe acne, Scalp fungal infection. Book an urgent GP referral or contact dermatology directly. Photograph the lesion to document any change.

**ROUTINE** — 1 to 3 months: Annual full-body skin check (for high-risk individuals); chronic eczema or psoriasis requiring new or adjusted treatment; hair loss evaluation; chronic urticaria management; rosacea not responding to first-line treatments; pigmentation concerns. Conditions to consider: Preventive cancer screening, Systemic dermatological review, Biologic candidacy assessment. Schedule a routine dermatology appointment; take photographs to the consultation; list all current medications and previous treatments tried.

**SELF-CARE** — OTC treatment appropriate (trial for 2–4 weeks): First-time mild athlete's foot; mild dandruff; mild sunburn without blistering; few blackheads or whiteheads on the face; minor insect bites; dry skin or xerosis; a single small wart. Conditions to consider: Tinea pedis, Pityriasis capitis, Mild acne, Xerosis, Verruca vulgaris. Use the appropriate over-the-counter treatment. See a doctor if there is no improvement in 2–4 weeks, if the condition is spreading, or if signs of infection develop (warmth, pus, or fever).

*MEDICAL DISCLAIMER*

*Professional medical consultation, diagnosis, or treatment. Always consult a qualified, licensed dermatologist or healthcare provider for any skin condition. E*